

Robot-as-a-Peer-in-Learning

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1. Introduction

Overview

Learners do not always use assistance just because it is available. In the rationale for this study, requesting help was treated as both an information-seeking action and a socially visible admission of uncertainty. A help control may therefore get used differently depending on what it returns, how often it can be activated, and whether the apparent source is positioned as a collaborator or an evaluator. The social role assigned to an educational agent is a plausible design variable, but it cannot be interpreted independently of the assistance mechanics through which that role is enacted.

Robots and computer agents can be made to occupy different positions in a learning relationship. A tutor is expected to possess knowledge, evaluate progress, and provide direction; a peer implies a more symmetrical relationship and a shared problem [Kanda et al., 2004]. Collaborative-learning theory describes relative symmetry, shared activity, and joint goals as important conditions rather than treating collaboration as simple co-presence [Dillenbourg, 1999]. In the present system, a “study buddy” label and first-person-plural language were intended to reduce hierarchy, whereas an “instructor” label presented the agent as an authority. Whether those cues actually changed behavior is an empirical question, not an assumption that the browser agent was experienced as human.

Educational-robot research gives this question some practical weight. A review of 101 studies containing 309 results reported cognitive and affective effects for robot tutoring of $d = 0.70$ and $d = 0.59$ respectively, alongside a cited benchmark of $d = 0.79$ for human one-to-one tutoring [Belpaeme et al., 2018]. Yet the roles represented in that literature were uneven: tutor roles accounted for 48% of results, teacher roles for 38%, and peer roles for only 9% [Belpaeme et al., 2018]. The prevalence of authority roles leaves the social and behavioral consequences of peer positioning comparatively uncertain, especially for adult learners [Salomons et al., 2022].

Previous studies suggest that robot role can influence engagement, social impressions, and learning, but much of that evidence concerns children, and the effects are not uniformly beneficial [Diyas et al., 2016, Kanda et al., 2004, Zaga et al., 2015]. Comparisons of a “friendly peer” with a “capable tutor” may also alter relationship and function together [Salomons et al., 2022]: if the peer supplies better hints or forgives errors, behavior can reflect either social framing or instrumental value.

This study contributes an adult, university-affiliated comparison in which help-button use was a primary outcome, as well as a methodological case study. The implementation

review showed that participants chose their own role, and that assistance, retries, scripts, reassurance, and modality differed between conditions. The analysis therefore has to separate the observed patterns from unsupported causal attribution.

Designers often express warmth through both language and generosity: a companion may speak collaboratively, give stronger assistance, and permit recovery from mistakes. That kind of bundle may suit a product, but it cannot isolate social equality on its own. The implemented bundle is therefore treated as the unit of comparison here, with scores and event logs used to diagnose task non-equivalence.

The study used a single-page Flask quiz with ten four-option European-football questions, drawn from one of two sets. A condition-specific introduction played through the Web Speech API. “Ask for Help” returned either progressive option elimination with collaborative text, or a fixed clue with instructor-oriented text. Client-side state locked the interaction during explanations, and two CSV files stored the session and event records.

The software was corrected after data collection. The current server route assigns peer or tutor using `random.choice` and passes that condition to the page. That code is appropriate for future sessions, but it does not change where the existing data came from — the 14 analysed observations came from an earlier build where people chose their own condition.

Research Questions

The questions retain during the study:

RQ1 — Help-seeking: Is assignment to the peer-labelled condition associated with more presses of “Ask for Help” than assignment to the tutor-labelled condition?

RQ2 — Comfort: Is assignment to the peer-labelled condition associated with higher post-task comfort than assignment to the tutor-labelled condition?

RQ3 — Performance diagnostic: Are quiz scores similar across conditions, as would be expected if the manipulation changes willingness to seek help and comfort without making one version of the task mechanically easier?

Hypotheses

The hypotheses retain during the study:

H1: Students who work with a robot peer will ask for help more often than students who work with a robot tutor.

H2: Students in the peer condition will report higher comfort, and implicitly lower stress, than students in the tutor condition, even if task scores are similar.

The hypotheses are described in the source materials as preregistered, which is also given as the reason for one-tailed tests. No registry identifier, public or archived URL, timestamp, or analysis plan is supplied, so the status of the preregistration — and the timing of the hypotheses, exclusions, test direction, and log inspection — is not independently verifiable. This report retains H1 and H2 as reported, preregistered hypotheses, without treating the registration claim itself as established.

2. Related Work

Educational robots can act as teachers, tutors, peers, or learners, with each role assigning expertise and responsibility differently [Belpaeme et al., 2018]. Although robot tutoring has shown positive educational effects, peer-based interactions remain less studied [Belpaeme et al., 2018]. Previous research has demonstrated that peer, tutor, and teacher roles can influence engagement and learning [Diyas et al., 2016, Kanda et al., 2004, Zaga et al., 2015]. However, role comparisons are only really interpretable when assistance and interaction rules are held constant.

In this study the “robot” was a browser-based agent rather than a physically present robot. Physical embodiment can affect human–robot interaction [Bainbridge et al., 2011], so the findings should be limited to screen-mediated role framing. The agent could present images, text, and synthesized speech, but it could not provide embodied cues such as mutual gaze, bodily orientation, or shared physical space.

The experiment contrasted a collaboratively framed study buddy with an instructor that provided clues. Although all participants correctly identified the intended classmate or teacher role, the manipulation also changed the assistance provided. Tutor requests revealed clues that required interpretation, whereas peer requests eliminated incorrect options and could eventually reveal the answer. Consequently, differences in help requests or performance cannot be attributed to role framing alone; they may just reflect the unequal usefulness of the assistance.

Prior knowledge is another important limitation. Previous research found that adults with low prior knowledge learned more from a peer robot than from a tutor robot, despite no overall difference in acquired skills [Salomons et al., 2022]. In the present study football knowledge was not measured, and participants selected their own condition, so baseline differences may have affected both assistance use and quiz performance.

Collaborative wording does not necessarily create fully collaborative learning [Dillenbourg, 1999], particularly when the participant answers individually and the agent has no independent responsibility. Similarly, the study does not test the protégé effect [Chase et al., 2009], since participants were not asked to teach the agent. Future experiments should hold hint content, retries, timing, modality, and task difficulty constant while varying only the relational framing, and should also measure prior knowledge and use graded, multi-item measures of perceived authority, peer-likeness, social presence, trust, and comfort.

3. System Design

3.1 Interaction flow

The phases that participants go through are shown across 6 stages in Figure 1 below.

Figure 2 shows the introductory screen in the corrected build, with the quiz instructions, “Start Quiz” control, and a server-assigned Peer role. The analysed data, though, came from the earlier build where participants selected their own role. Figure 2 explicitly states that the companion is randomly assigned — that statement describes the corrected software only, and it must not be read as evidence that the 14 analysed sessions were randomized.

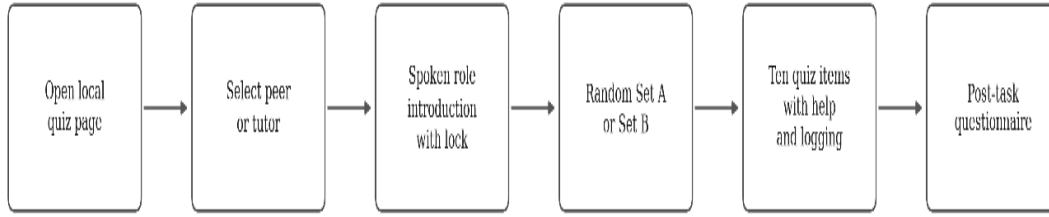


Figure 1: Overview of the study procedure completed by each participant.

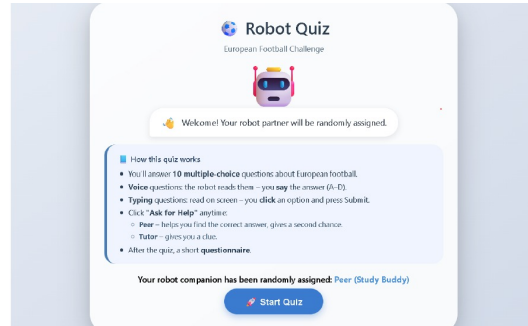


Figure 2: Introduction screen.

Figure 3 shows the session quiz screen with a European-football item, four options, a voice-mode indicator, submission control, help control, and a speech-error notice. The visual shell was shared across conditions, but assistance, recovery, scripts, and modality balance were not equivalent. This page is considered the next page after the introductory screen.

3.2 Architecture and data flow

The prototype consisted of a Flask server in `app.py` and a single-page client in `templates/index.html`. The server supplied quiz content, evaluated answers, selected condition-dependent assistance, and recorded events, while the browser rendered the interface, played speech, controlled the explanation lock, and updated the available options. This division kept the visible interaction responsive while retaining session records server-side.

Two CSV files stored the study output. `experiment_data.csv` contained session-level data such as condition, score, and questionnaire responses. `help_log.csv` retained question-level requests and related events, which allowed repeated help use to be examined rather than just inferred from the total. This event log was essential for identifying the 30-request pattern of one participant.

After data collection, the root route was revised to execute `random.choice(['peer', 'tutor'])` and pass the result into the template. The page now reports an assigned “Peer (Study Buddy)” or “Tutor (Instructor).” This change corrects future allocation, but it has no retrospective effect on the dataset already collected.

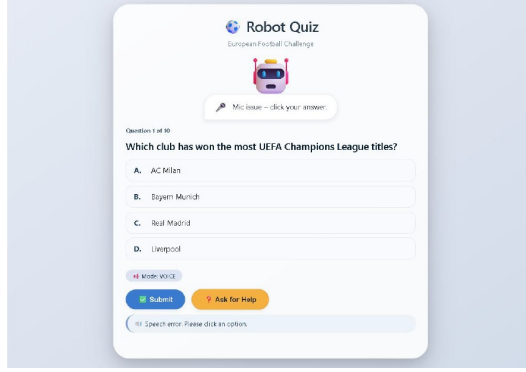


Figure 3: Quiz question screen.

3.3 Content and implementation decisions

The quiz covered European football. Two sets, each with ten four-option items, were used; example topics included the club with the most UEFA Champions League titles and the leading scorer at the 2022 World Cup. The choice of trivia was meant to avoid assessing material from a specific university module and to keep the roughly 20-minute session less exam-like. It did not, however, ensure equivalent prior knowledge — familiarity with football can vary a lot, and no baseline test or self-rating was collected.

Each item had a short, pre-authored factual clue. In the final implementation that clue was used in the tutor condition, while the peer condition acted on the choice set instead. Before this fixed-response design, a local DistilGPT-2 prototype was tested through Hugging Face transformers. It was discontinued because responses took roughly three to five seconds, the generated text could invent or misstate facts, local inference put substantial CPU and memory demands on the laptops, and open-ended outputs were hard to audit. Fixed templates returned in under 100 milliseconds, constrained the factual content, and allowed help events to be traced. This decision improved speed and auditability, but using different kinds of fixed assistance in the two conditions created a separate validity problem.

3.4 Condition comparison between peer and tutor

As Table 1 shows, with four choices, three peer requests can remove all three wrong alternatives. The help button can thus convert uncertainty into certainty without requiring any domain knowledge. The tutor clue may be useful, but it does not guarantee convergence, and repeated activation does not provide an equivalent sequence of new eliminations. The peer-only retry further lowers the cost of an initial mistake. Accordingly, the delivered contrast combined role language, assistance utility, repeat value, revision opportunity, script content, reassurance, and modality — it was not a controlled manipulation of peer versus tutor status alone.

Feature	Peer condition	Tutor condition	Consequence for interpretation
Social framing	“Study Buddy”; informal collaborative language using “we” and “us”	“Instructor”; more formal and authoritative language	Intended relational manipulation
Introduction	Approximately 110 words; more reassuring; explicitly previews a second chance	Approximately 75 words; mentions clue availability	Length, tone, reassurance, and disclosed affordances differ
Help response	Randomly removes one remaining incorrect option, crosses it out, and selects from ten collaborative templates	Presents the item’s fixed factual clue using one of four framing lines; no option disappears	Information and mechanical effect are unequal
Repeated help	Each request can remove another wrong option; no reported limit	Repetition does not progressively narrow the answer set	A request has different marginal value across conditions
Error recovery	One retry after the first wrong answer	No retry	Final-answer opportunity and cost of guessing differ
Input modality	More typed questions	More voice-based questions	Modality is entangled with role
Visible layout	Common quiz layout and controls	Common quiz layout and controls	Visual constancy does not equal procedural constancy

Table 1: Comparison of peer and tutor conditions.

4. Methods

4.1 Participants

Fourteen university-affiliated adults took part. Ages ranged from 23 to 26; 12 participants were men and two were women. Seven participants entered the peer condition and seven entered the tutor condition. Recruitment was based on convenience, and sessions were conducted mostly with people personally known to the researcher. The sample was smaller than the 20–24 participants (or 10–12 per condition) called for in an ethics-application draft, but larger than an earlier six-person pilot concept. No power analysis, sampling frame, recruitment rate, or compensation information is reported. The narrow, personally recruited, male-skewed university sample constrains generalization and does not support a gender comparison.

Design and allocation

The intended study was a two-condition, between-subjects comparison. A between-subjects structure avoided carry-over from exposing one participant to both the casual companion and the instructor-styled agent. Planning materials referred to allocation by coin flip or another random procedure.

The executed procedure differed. In the analysed build, participants clicked a role and thereby selected their own condition. Equal group sizes do not turn that process into random allocation. Selection could be related to pre-existing role preference, comfort with authority, willingness to request assistance, confidence, football knowledge, or a preference for speaking versus typing. Those possible baseline differences were not measured, so the reported between-group tests cannot adjust for them. The server-side `random.choice` code now present in the application is prospective only.

Materials and procedure

Materials comprised the Flask quiz, the two football question sets, condition-specific spoken introductions, automated logs, an information sheet, consent materials for sessions conducted under the ethics protocol, and a short post-task questionnaire.

Participants read the information sheet and, where applicable under the protocol, signed consent. They then selected peer or tutor in the data-collection interface, heard the corresponding welcome and instructions, and learned that help was available during the task. One of the two ten-item sets was selected randomly for the session. Each question appeared separately, and the application recorded help requests, retries where the interface permitted them, and final answers. After all items, participants responded to the comfort item, indicated whether the robot had acted more like a classmate or a teacher, supplied age and gender, and were thanked. Sessions were conducted remotely through a locally hosted server and lasted about 20 minutes.

Measures

Construct	Operational measure	Scale	Analytical role
Help-seeking	Total presses of “Ask for Help” across ten items	Count	Primary outcome for H1 and RQ1
Comfort	One post-task rating	1–7	Primary outcome for H2 and RQ2
Performance	Number of final answers correct	0–10	Diagnostic outcome for RQ3
Perceived role	“Did the robot act more like a classmate or more like a teacher?”	Forced choice	Manipulation check

Table 2: Dependent variables and their operational measures.

Table 2 lists the three dependent variables used in this study: help-request count, comfort scale (1–7), and task score (0–10). The planning draft referred to percentage

correct, whereas the application stored a raw count. Since all participants answered ten questions, one correct answer equals ten percentage points, so the two representations are linear transformations and do not change the standardized group comparison.

The measures have important boundaries. Total button presses are observable, but “help-seeking” is only a valid interpretation when a press has comparable purpose and pay-off, and the different assistance mechanisms violate that condition. The comfort outcome consists of one unvalidated item (Table 2), so internal reliability cannot be estimated, and the rating cannot distinguish interpersonal ease from satisfaction, confidence, task ease, relief, anxiety, or positive affect after a good score. The role check establishes a binary classification (Table 2), not the strength or dimensions of perceived status.

No established multi-item HRI measure was administered. Future work should select and validate a construct-specific battery before data collection, but no such measure is available for retrospective secondary analysis here.

Analysis

The reported analysis used independent-samples t tests implemented with `scipy.stats.ttest_ind` and $\alpha = .05$. H1 and H2 were evaluated with one-tailed tests in the predicted direction, peer greater than tutor. Performance was evaluated with a two-tailed test since no directional score difference was predicted. Cohen’s d was reported for each contrast, and manipulation-check counts and the event-level entries in `help_log.csv` were also inspected. No new inferential test is introduced in this revision; no participant is removed, no nonparametric replacement is presented, and no covariate adjustment is possible. With seven observations per group, distributional assumptions are hard to assess, and a highly dispersed count can be strongly influenced by a single person. Event-log interpretation is diagnostically valuable but was not documented as a preregistered inferential analysis, and as noted above, the preregistration claim itself cannot be verified from a registry record.

5. Results



Figure 4: Help requests, comfort ratings, and task scores by condition.

Figure 4 shows the individual observations and condition means for help requests, comfort, and quiz score. The peer group recorded a higher mean number of help requests

than the tutor group: $M = 12.14$, $SD = 11.84$, compared with $M = 6.00$, $SD = 3.21$ (Figure 4). The reported one-tailed test was $t(12) = 1.325$, $p = .105$, with Cohen’s $d = 0.71$. At $\alpha = .05$, H1 was not supported. The peer distribution ranged from 0 to 30 requests, and two peer participants made no requests at all; tutor counts ranged from 0 to 11. As is visible in Figure 4, one participant in the peer condition accumulated 30 requests — frequently using three on a single question — and still obtained a score of 10/10. So the peer mean came with substantial within-condition heterogeneity.

Post-task comfort ratings showed the pattern predicted by H2 (Figure 4). The peer group’s mean was $M = 5.71$, $SD = 0.95$, against a tutor mean of $M = 3.57$, $SD = 0.79$. The directional comparison was $t(12) = 4.593$, $p < .001$, with $d = 2.46$. The observed ordering matched H2’s prediction, and the standardized difference was large in this sample.

Contrary to the expectation that task performance would be similar across conditions, quiz scores were higher in the peer condition, $M = 7.29$, $SD = 2.36$, than in the tutor condition, $M = 3.57$, $SD = 1.72$ (Figure 4). The reported two-tailed test was $t(12) = 3.366$, $p = .006$, with $d = 1.80$.

Role Identification

All 14 participants chose the response consistent with their condition. The seven peer participants described the agent as more like a classmate, and the seven tutor participants described it as more like a teacher. No participant was excluded for failing the check.

The printed output from `analysis.py` contained a labeling error: the tutor failure count was printed twice, once for each condition row. Inspection of the grouped data nevertheless confirmed seven passes in each condition and a 100% recognition rate. The bug affects the printed label, not the grouped conclusion.

Statistical observations

Outcome	Peer, $M \pm SD$	Tutor, $M \pm SD$	Reported comparison	Statistical decision
Help requests	12.14 $\pm$ 11.84	6.00 $\pm$ 3.21	$t(12) = 1.325$, one-tailed $p = .105$, $d = 0.71$	H1 not supported at $\alpha = .05$
Comfort (1–7)	5.71 $\pm$ 0.95	3.57 $\pm$ 0.79	$t(12) = 4.593$, one-tailed $p < .001$, $d = 2.46$	Predicted ordering observed
Score (0–10)	7.29 $\pm$ 2.36	3.57 $\pm$ 1.72	$t(12) = 3.366$, two-tailed $p = .006$, $d = 1.80$	Unexpected inequality

Table 3: Condition means, standard deviations, test statistics, and statistical decisions for all three outcomes.

Table 3 summarizes the condition means, standard deviations, test statistics, and statistical decisions for all three outcomes.

Beyond the group comparisons, the assistance design itself is worth a diagnostic note. Repeated help in the peer condition had a direct functional relation to answer availability: on a four-option item, three eliminations could leave only the correct choice. That same

sequence was not available in the tutor condition, where help provided only a clue requiring interpretation. This is reported as a diagnostic property of the interface rather than as an additional hypothesis test.

6. Discussion and Conclusion

Help-seeking

The peer group pressed “Ask for Help” about twice as often on average as the tutor group, but the one-tailed test did not reach $\alpha = .05$, so H1 was not supported. The result is best described as inconclusive rather than null. The peer distribution ranged from 0 to 30 requests, two peer participants never asked for help, and a single participant who requested help on almost every option accounted for much of the group mean — with only seven observations per group, one case like that dominates both the mean and the variance. More fundamentally, a request had a different payoff in each condition. In the peer condition it guaranteed removal of one wrong option and stayed useful when repeated, so three presses on a four-option item could leave only the correct answer. In the tutor condition it returned a fixed clue whose value depended on prior knowledge and did not increase with repetition. Total request counts therefore mix social willingness to disclose uncertainty with a rational response to unequal information, and the present data cannot separate the two. What can be said is that the frequency of help use tracked the utility of help, which is a design finding rather than a test of peer framing.

Comfort

Post-task comfort was substantially higher in the peer condition, the predicted ordering under H2 was observed, and the standardized difference was large. This is the strongest statistical result in the study, yet it only supports a condition-level association in this sample. Participants who preferred an informal helper may have selected the peer role and rated it more comfortably for that reason. The peer experience was also more forgiving in every respect that plausibly affects comfort: it offered a retry, announced that safety net in a longer and more reassuring introduction, delivered more decisive help, involved more typed rather than spoken input, and ended with higher scores. A single unvalidated item cannot tell us whether the rating reflected interpersonal ease, reduced anxiety, satisfaction with success, or relief at task ease. The comfort difference is thus consistent with the idea that collaborative framing lowers the perceived hierarchy of asking, but it is equally consistent with several non-relational explanations that the design did not control for.

Performance

RQ3 was framed as a check on equivalence: if the manipulation changed only willingness to seek help and comfort, scores should have been similar. They were not. The peer group scored markedly higher, and the two-tailed difference was significant with a large effect. This is the most informative outcome, because it shows the conditions were not functionally equivalent tasks. Progressive option elimination and the peer-only retry

provided direct routes to a correct final answer that the tutor version lacked, and unmeasured football knowledge together with self-selection remain additional explanations. The score difference should not be read as evidence that a peer robot improves learning — no learning gain or baseline was measured, and the peer version simply made correct answers easier to obtain. Instead, the failed similarity diagnostic retrospectively qualifies RQ1 and RQ2: whatever produced the score gap could also have produced more help use and higher comfort.

Interpretation

Taken together, the answers form a coherent pattern that points to the assistance design rather than to social role. Help was used more where it paid more, comfort was higher where the task was more forgiving and success more frequent, and scores were higher where the interface could reveal answers. The role-recognition result confirms that participants noticed the classmate/teacher distinction, but recognition does not isolate cause. The findings agree with prior reviews in showing that affective and cognitive outcomes should be reported separately [Belpaeme et al., 2018], and with earlier work in flagging prior domain knowledge as a plausible moderator that was not measured here [Salomons et al., 2022]. They also remain confined to a screen-mediated agent; prior research shows that physical presence changes interaction [Bainbridge et al., 2011], so extension to a co-present robot is a separate question.

Conclusion

In this 14-person sample, the implemented peer agent was associated with higher comfort and higher quiz scores, while its higher mean number of help requests did not reach the reported significance criterion. H1 was not supported, H2’s predicted ordering was observed, and the expected similarity in scores was not. All participants identified the intended role.

These results do not demonstrate that being helped by a robot peer caused any of the differences. Participants selected their own condition, and the peer version bundled collaborative language with more informative and repeatable help, a retry, a longer and more reassuring introduction, and a different input modality, while prior knowledge went unmeasured. The failed performance diagnostic indicates that the two conditions differed as tasks, not only as relationships.

The study’s defensible contribution is therefore methodological. Social role in educational human–robot interaction can only be studied when relational framing is separated from capability. Random assignment, equal hint information, equal request limits and retry rules, matched scripts, controlled modality, a baseline knowledge measure, and multi-item outcome scales are prerequisites before comfort or help behavior can be attributed to peer rather than tutor framing. The present prototype and its logs show pretty concretely why these controls matter.

7. Limitations

The most serious limitation is that participants selected their own condition, so causal identification is not possible. Role preference, confidence, willingness to ask for help, and other unmeasured baseline differences could have preceded the outcomes, and the randomization code now in the application cannot alter the data already collected. This problem is compounded by the small and narrow sample: 14 university-affiliated adults aged 23 to 26, seven per group, mostly men and mostly known to the researcher — estimates here are imprecise, sensitive to individual cases, and below the planned 20 to 24 participants.

The two conditions also differed in more than just role framing. Peer help guaranteed removal of a wrong option and remained useful when repeated, whereas tutor help exposed a factual clue without progressive narrowing, so the behavior counted as help-seeking had a different payoff in each condition, and repeated requests could function as an answer-finding procedure rather than ordinary help-seeking. Only peer participants could retry after an error, the peer introduction was longer and more reassuring, and input modality was linked to condition. Prior football knowledge, which plausibly interacts with all of these features, was not measured or balanced.

Finally, the measures were limited. Comfort was captured with a single unvalidated item, so its reliability cannot be estimated, and it cannot distinguish interpersonal ease from satisfaction, confidence, or relief after success. The manipulation check established only categorical recognition of the classmate or teacher role, not its strength or dimensions. In addition, the agent was a screen-based interface rather than a physically present robot [Bainbridge et al., 2011], and the preregistration of the hypotheses cannot be independently verified, which qualifies the confirmatory status of the one-tailed tests.

8. Future Work

The first priority is a controlled replication with concealed server-side random allocation, a sample of at least the planned 20 to 24 participants recruited beyond personal contacts and ideally sized by an a priori power analysis, and conditions that differ only in relational wording: both groups should receive the same assistance mechanism with matched request limits, timing, and retry rules, introductory scripts of comparable length and content, input modality standardized or randomized independently of role, and the two question sets balanced across conditions and checked for comparable difficulty. A factorial design could further separate peer versus tutor language from the type of assistance offered. Measurement should be strengthened by assessing prior football knowledge before exposure to role materials, since previous research found that initial domain knowledge moderated adult outcomes in a peer–tutor robot comparison [Salomons et al., 2022]; by replacing the single comfort item with a prespecified multi-item instrument covering constructs such as anxiety, social presence, trust, and satisfaction; by using graded ratings of peer-likeness and teacher-likeness instead of a forced choice; and by decomposing help-seeking into meaningful events such as first requests, repeated requests, and requests after errors, with the analysis of counts, outliers, and exclusions specified in advance. Once role framing has been isolated in the browser, later studies could examine adaptive systems that shift between roles according to learner state, and could cross role framing

with physical embodiment, which has been shown to alter interaction [Bainbridge et al., 2011], while preserving the same controls. Any future preregistration should name the registry, identifier, hypotheses, primary outcomes, test direction, and exclusion rules, so confirmatory and diagnostic analyses can be clearly distinguished.

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AI-Use Declaration

The source report states that ChatGPT and Claude were used to improve grammar, clarity, LaTeX formatting, and LaTeX code, while responsibility for the study concept, design, data collection, analysis, results, and conclusions remained with the author. It identifies the following Claude conversation in relation to LaTeX assistance: <https://claude.ai/chat/56a5bd1b-fd17-48ec-b7a9-16a4af73df11>.